

Moulds & Dies Technology

□ To provide a thorough knowledge on plastic materials, their properties and processing methods in moulding industry

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3111 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3111.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Tool and Die Engineering
Course code	3111
Course title	Moulds & Dies Technology
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To provide a thorough knowledge on plastic materials, their properties and processing methods in moulding industry
- □ To introduce different plastic processing techniques used in industry
- □ To give a basic information on the product design concept

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding processing methods in moulding industry	See official syllabus
CO2	Apply feed and cooling system in mould 15 Applying Explain the methods of plastic moulding other than	See official syllabus
CO3	14 Understanding injection moulding Recognize process description of die casting and	See official syllabus
CO4	14 Understanding the concept of product design	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3 2
CO3	CO3 2 1
CO4	CO4 2 1

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	industry	4	As specified
Module 2	Apply feed and cooling systems in mould	4	As specified
Module 3	process	4	As specified
Module 4	product design	4	As specified

MODULE 1

industry

This module is presented from the official syllabus scope for Moulds & Dies Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: industry
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Classify plastic materials and their properties 3 Understanding Differentiate between thermo and

Classify plastic materials and their properties 3 Understanding Differentiate between thermo and is an official topic in Module 1 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify plastic materials and their properties 3 Understanding Differentiate between thermo and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify plastic materials and their properties 3 understanding differentiate between thermo and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify plastic materials and their properties 3 Understanding Differentiate between thermo and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

4 Understanding thermosetting plastics Explain various moulding processes used in

4 Understanding thermosetting plastics Explain various moulding processes used in is an official topic in Module 1 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding thermosetting plastics Explain various moulding processes used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding thermosetting plastics explain various moulding processes used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding thermosetting plastics Explain various moulding processes used in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding thermosetting plastics Explain various moulding processes used in പ്ലാസ്റ്റിക് മോൾഡിംഗ് definition/meaning പ്ലാസ്റ്റിക് മോൾഡിംഗ്, steps, application പ്ലാസ്റ്റിക് മോൾഡിംഗ് പ്ലാസ്റ്റിക്. Technical terms English-പ്ലാസ്റ്റിക് മോൾഡിംഗ്.

4 Understanding Industry Describe working principle of injection

4 Understanding Industry Describe working principle of injection is an official topic in Module 1 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Industry Describe working principle of injection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding industry describe working principle of injection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Industry Describe working principle of injection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-4 Understanding Industry Describe working principle of injection മലയാളം definition/meaning മലയാളം. മലയാളം മലയാളം മലയാളം, steps, application മലയാളം മലയാളം മലയാളം. Technical terms English- മലയാളം മലയാളം മലയാളം.

4 Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding-working-principle-advantages and disadvantages

4 Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding- working-principle-advantages and disadvantages is an official topic in Module 1 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding- working-principle-advantages and disadvantages using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding moulding machine plastics in the mould industry - definition of plastics - properties of plastics - application. - types thermo setting plastics - thermo plastics - examples - properties and uses of nylon (polyamide),pvc, pmma, acrylic, polycarbonate, frp, polyoxymethylene (pom), polystyrene (ps), abs, polypropylene (pp),polyethylene (pe) plastic processes: injection moulding, compression moulding, transfer moulding; extruding; casting; calendaring; fabrication methods-sheet forming, blow moulding, laminating plastics (sheets, rods & tubes) injection moulding- working-principle-advantages and disadvantages should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding- working-principle-advantages and disadvantages means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 4 Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding- working-

principle-advantages and disadvantages ഏകദേശം definition/meaning
ഏകദേശം. ഏകദേശം ഏകദേശം, steps, application ഏകദേശം
ഏകദേശം. Technical terms English-ഏകദേശം ഏകദേശം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

industry		
M1.01	Classify plastic materials and their properties	3
Understanding	Differentiate between thermo and	
M1.02		4
Understanding	thermosetting plastics	
	Explain various moulding processes used in	
M1.03		4
Understanding	Industry	
	Describe working principle of injection	
M1.04		4
Understanding	moulding machine	
Contents:		
Plastics in the mould industry - Definition of plastics - properties of plastics		
- application. -		
types		
Thermo setting plastics - Thermo plastics - Examples - properties and uses of		
Nylon		
(Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM),		
Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE)		
Plastic processes: Injection moulding, Compression moulding, Transfer moulding;		
Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow		
moulding,		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Apply feed and cooling systems in mould

This module is presented from the official syllabus scope for Moulds & Dies Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply feed and cooling systems in mould
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Terminology used in mould making

Terminology used in mould making is an official topic in Module 2 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Terminology used in mould making using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, terminology used in mould making should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Terminology used in mould making means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Terminology used in mould making

000000000000 00000 definition/meaning 000000000000. 000000 000000 00000000,
 steps, application 000000 0000000000 000000. Technical terms English-0 00000
 000000000000.

Apply Feed system of mould

Apply Feed system of mould is an official topic in Module 2 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply Feed system of mould using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply feed system of mould should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply Feed system of mould means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എപ്പ്ലി ഫീഡ് സിസ്റ്റം ഓഫ് മൗൾഡ് $\square\square\square\square\square\square\square\square\square$
 $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps,
application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square\square$
 $\square\square\square\square\square\square\square\square\square$.

Explain Ejection techniques 4 Understanding Apply cooling system and classification of

Explain Ejection techniques 4 Understanding Apply cooling system and classification of is an official topic in Module 2 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Ejection techniques 4 Understanding Apply cooling system and classification of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain ejection techniques 4 understanding apply cooling system and classification of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Ejection techniques 4 Understanding Apply cooling system and classification of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain Ejection techniques 4 Understanding Apply cooling system and classification of ഏകദേശം definition/meaning ഉപയോഗിക്കുന്നു. എങ്ങനെ ഉപയോഗിക്കുന്നു, steps, application ഉപയോഗിക്കുന്നു ഉപയോഗിക്കുന്നു. Technical terms English-എ ഉപയോഗിക്കുന്നു.

4 Applying mould Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding

4 Applying mould Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding is an official topic in Module 2 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying mould Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying mould definition of mould -terminology - mould materials feed system - runner - types of runner - runner diameter calculation - gate - types of gates - gate calculation ejection techniques - parting surface - venting - shrinkage - values of shrinkage - factors governing shrinkage temperature control - cooling - mould cooling (simple numerical problem) side core and side cavity - determine core and cavity dimensions - internal undercuts - two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. explain plastic processing techniques other than injection moulding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying mould Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 4 Applying mould Definition of mould - Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding definition/meaning.

താഴെ പറയുന്നവയെക്കുറിച്ച്, steps, application തുടങ്ങിയവയെക്കുറിച്ച്. Technical terms English-ൽ എഴുതുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Apply feed and cooling systems in mould

M2.01	Terminology used in mould making	3
Understanding		

M2.02	Apply Feed system of mould	4
Applying		

M2.03	Explain Ejection techniques	4
Understanding		

M2.04	Apply cooling system and classification of	4
Applying		

mould		
Series Test - I		1

Contents:

Definition of mould -Terminology - mould materials Feed system - Runner - types of

runner - Runner diameter calculation - Gate - types of gates - Gate calculation

Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors

governing shrinkage

Temperature control - cooling - mould cooling (simple numerical problem)

Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two

plate and three plate moulds - runner less moulds - multi daylight moulds - mould

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

process

This module is presented from the official syllabus scope for Moulds & Dies Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: process
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding moulding Explain process of extrusion and blow

2 Understanding moulding Explain process of extrusion and blow is an official topic in Module 3 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding moulding Explain process of extrusion and blow using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding moulding explain process of extrusion and blow should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding moulding Explain process of extrusion and blow means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 2 Understanding moulding Explain process of extrusion and blow definition/meaning steps, application Technical terms English-

4 Understanding moulding

4 Understanding moulding is an official topic in Module 3 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding moulding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding moulding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding moulding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum

Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum is an official topic in Module 3 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the process of rotational moulding 4 understanding describe the process of calendaring, vacuum should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എം Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum എംഎംഎംഎംഎംഎംഎംഎംഎം definition/meaning എംഎംഎംഎംഎംഎംഎം. എംഎംഎം എംഎംഎം എംഎംഎംഎം, steps, application എംഎംഎം എംഎംഎംഎംഎം എംഎംഎം. Technical terms English-എം എംഎംഎം എംഎംഎംഎംഎംഎം.

4 Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming - filament winding - multicolor moulding Recognize process description of die casting and introduce the concept of

4 Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming - filament winding - multicolor moulding Recognize process description of die casting and introduce the concept of is an official topic in Module 3 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming - filament winding - multicolor moulding Recognize process description of die casting and introduce the concept of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding forming compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type extrusion moulding - description of the process - application. blow moulding - description of the process - application. rotational moulding - description of the process - application. calendaring - vacuum forming - filament winding - multicolor moulding recognize process description of die casting and introduce the concept of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming - filament winding - multicolor moulding Recognize process description of die casting and introduce the concept of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming - filament winding - multicolor moulding Recognize

process description of die casting and introduce the concept of **process**
process definition/meaning **process**. **process** **process** **process**, steps,
application **process** **process** **process**. Technical terms English-**process**
process.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

process

Explain process of compression and transfer

M3.01

2

Understanding

moulding

Explain process of extrusion and blow

M3.02

4

Understanding

moulding

M3.03

4

Understanding

Describe the process of calendaring, vacuum

M3.04

4

Understanding

forming

Contents:

Compression moulding - application - temperature and pressure ranges - basic types of

compression moulds - flash type mould - positive mould - landed positive mould - semi

positive mould.

Transfer moulding - application - advantages and limitations - types of transfer mould - pot

type and plunger type

Extrusion moulding - description of the process - application.

Blow moulding - Description of the process - application.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

product design

This module is presented from the official syllabus scope for Moulds & Dies Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: product design
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain die casting process 4 Understanding Classify Die casting machines - hot chamber

Explain die casting process 4 Understanding Classify Die casting machines - hot chamber is an official topic in Module 4 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain die casting process 4 Understanding Classify Die casting machines - hot chamber using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain die casting process 4 understanding classify die casting machines - hot chamber should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain die casting process 4 Understanding Classify Die casting machines - hot chamber means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിരിക്കട്ടെ die casting process 4 Understanding Classify Die casting machines - hot chamber എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ definition/meaning എന്നിരിക്കട്ടെ. എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ, steps, application എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ. Technical terms English-എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ എന്നിരിക്കട്ടെ.

4 Understanding and cold chamber

4 Understanding and cold chamber is an official topic in Module 4 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and cold chamber using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and cold chamber should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and cold chamber means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding and cold chamber

definition/meaning . . . ,
steps, application Technical terms English-
.

List Die casting alloys

List Die casting alloys is an official topic in Module 4 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List Die casting alloys using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list die casting alloys should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List Die casting alloys means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- List Die casting alloys ഏകദേശം 1000 definition/meaning ഉൾപ്പെടെ. ഉൾപ്പെടെ 1000 steps, application ഉൾപ്പെടെ 1000 Technical terms English- ഉൾപ്പെടെ 1000

Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: Sl.No Website Link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/>

Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: Sl.No Website Link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/> is an official topic in Module 4 of Moulds & Dies Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: Sl.No Website Link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the basic concept of product design 4 understanding die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process text/ reference: t/r book title/author t1 introduction to injection moulding - pye t2 the mould design guide - peter jones r1 fundamentals of plastics mould design - sanjay k nayak - tata mac graw product design and development-karl t. ulrich ,steven d. eppinger-mac graw r2 hill r3 plastic technology handbook - donald e hudgin online resources: sl.no website link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: Sl.No Website Link 1 https://myplasticmold.com/ 2 https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection- 3 https://plasticmold.net/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: Sl.No Website Link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/> definition/meaning . steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

product design		
M4.01	Explain die casting process	4
Understanding		
	Classify Die casting machines - hot chamber	
M4.02		4
Understanding		
	and cold chamber	
M4.03	List Die casting alloys	2
Understanding		
M4.04	Explain the basic concept of Product design	4
Understanding		
	Series Test - II	1

Contents:

Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies
Product design - introduction - steps involved in product design - characteristics of product

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Classify plastic materials and their properties 3 Understanding Differentiate between thermo and. (3 marks)

Answer: Begin with the standard definition or identity of *Classify plastic materials and their properties 3 Understanding Differentiate between thermo and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding thermosetting plastics Explain various moulding processes used in. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding thermosetting plastics Explain various moulding processes used in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding Industry Describe working principle of injection. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding Industry Describe working principle of injection*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide),PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP),Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding-working-principle-advantages and disadvantages. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding moulding machine Plastics in the mould industry - Definition of plastics - properties of plastics -*

application. - types Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide), PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP), Polyethylene (PE) Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes) Injection moulding- working-principle- advantages and disadvantages. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: താഴെ പറയുന്നവയുടെ നിർവ്വചനം, പ്രവർത്തന തത്വം/ഘട്ടങ്ങൾ, പ്രാധാന്യം എന്നിവ ഉൾക്കൊള്ളുന്ന ഒരു പ്രസ്താവന തയ്യാറാക്കുക.

Module 2

Define or explain Terminology used in mould making. (3 marks)

Answer: Begin with the standard definition or identity of *Terminology used in mould making*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: താഴെ പറയുന്നവയുടെ നിർവ്വചനം, പ്രവർത്തന തത്വം/ഘട്ടങ്ങൾ, പ്രാധാന്യം എന്നിവ ഉൾക്കൊള്ളുന്ന ഒരു പ്രസ്താവന തയ്യാറാക്കുക.

Define or explain Apply Feed system of mould. (3 marks)

Answer: Begin with the standard definition or identity of *Apply Feed system of mould*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: താഴെ പറയുന്നവയുടെ നിർവ്വചനം, പ്രവർത്തന തത്വം/ഘട്ടങ്ങൾ, പ്രാധാന്യം എന്നിവ ഉൾക്കൊള്ളുന്ന ഒരു പ്രസ്താവന തയ്യാറാക്കുക.

Define or explain Explain Ejection techniques 4 Understanding Apply cooling system and classification of. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Ejection techniques* 4 *Understanding Apply cooling system and classification of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying mould Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying mould Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance. Explain plastic processing techniques other than injection moulding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding moulding Explain process of extrusion and blow. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding moulding* Explain process of extrusion and blow. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding moulding. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding moulding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the process of rotational moulding 4 Understanding Describe the process of calendaring, vacuum*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process -

application. calendaring - vacuum forming -filament winding - multicolor moulding Recognize process description of die casting and introduce the concept of. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding forming Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming -filament winding - multicolor moulding Recognize process description of die casting and introduce the concept of.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain die casting process 4 Understanding Classify Die casting machines - hot chamber. (3 marks)

Answer: Begin with the standard definition or identity of *Explain die casting process 4 Understanding Classify Die casting machines - hot chamber.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding and cold chamber. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding and cold chamber.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain List Die casting alloys. (3 marks)

Answer: Begin with the standard definition or identity of *List Die casting alloys*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process Text/ Reference: T/R Book Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: SI.No Website Link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/>. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the basic concept of Product design 4 Understanding Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process* Text/ Reference: T/R Book

Title/Author T1 Introduction to injection moulding - Pye T2 The Mould Design Guide - Peter Jones R1 Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw R2 Hill R3 Plastic technology handbook - Donald E Hudgin Online Resources: Sl.No Website Link 1 <https://myplasticmold.com/> 2 <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-> 3 <https://plasticmold.net/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Classify plastic materials and their properties** 3 Understanding Differentiate between thermo and.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Terminology used in mould making**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding moulding Explain process of extrusion and blow.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain die casting process 4 Understanding Classify Die casting machines - hot chamber.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.

- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://myplasticmold.com/>
- <https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-https://plasticmold.net/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3111. Verify institutional updates against the current official curriculum.